1. TITLE OF THE IDEA

Apex PdM: Township-Scale Asset Health & Air-Conditioning Predictive Maintenance Platform

Eliminating Parasitic Energy Waste & Premature Landfill Scrap from Scaled Testbed to Commercial Infrastructure

`LIVE WORKING RIG: https://washer-pdm.pages.dev` **TRL Level:** TRL 5/6 (Functional Hardware Rig & Live Cloud)

15%–35%	< 1 Month	+5–10 Yrs	< RM 50
PARASITIC FRICTION CUT	COMMERCIAL PAYBACK	LIFESPAN EXTENSION	EDGE POD COST

2. PROBLEM STATEMENT

What is the issue? Mechanical rotating equipment (commercial central air-conditioning chillers, water pumps, large appliances) is managed reactively through "run-to-failure." When consumable wear parts (bearings, dampers, mechanical seals) degrade, internal friction forces inverter motors to draw **15% to 35% excess electrical power (W)** to maintain operational speed—undetected by circuit breakers for months. Eventually, unchecked friction causes catastrophic seizure (motor burnout, compressor seizure), turning an **RM 25–50 component fix** into an **RM 800–RM 1,800+ replacement** for appliances, or **RM 15,000–RM 30,000+** for commercial HVAC chillers.

Who is affected?

- **Township Communal & Commercial Spaces:** Community **mosques, suraus, kuils (temples), clubhouses, and retail shopping malls** operating 20 to 50+ commercial AC compressors daily. Sudden breakdown leaves congregations and shoppers in unbearable tropical heat.
- **Township Facility Management (FM) Teams:** Technicians waste hundreds of hours on manual 1-by-1 physical walk-throughs, unable to detect internal mechanical degradation before catastrophic breakdown occurs.
- **Township Residents:** Urban households facing expensive appliance repairs, structural noise, and inflated utility bills.

Why is it important to solve this problem now? Rising commercial electricity tariffs (TNB Tariff B @ RM 0.509/kWh) and national ESG circular economy mandates demand an end to preventable waste. Eliminating unmonitored mechanical friction saves gigawatt-hours across township utility grids, while intervening early at **Point P on the P-F Interval** extends machinery lifespan by 5 to 10 years, diverting metric tons of heavy scrap metal and appliance e-waste from landfills.

3. PROPOSED SOLUTION

How does your solution address the problem? Apex PdM provides low-cost, non-intrusive edge-IoT sensor pods that clip onto rotating machinery housings. It continuously tracks a **Machine Health Index (0%–100%)** and **Remaining Useful Life (RUL)**, detecting mechanical friction weeks in advance (Point P) before catastrophic failure (Point F).

1. Kinematics (MPU-6050 @ 50 Hz): Tri-axial digital acceleration (±16g) capturing structural unbalance, rocking, and damper fatigue.

2. Acoustics (INMP441 @ 16 kHz): 24-bit digital I2S DMA sound pressure detecting high-frequency bearing squeal and pump cavitation.

3. Electrical Power (Tapo P110 @ 1 Hz): Active wattage (W), voltage, current, and parasitic motor drag.

4. Operational Context: Conditions anomaly detection against machine states, eliminating spin false alarms.

What technology / product is involved?

- **Dual-Core ESP32 Edge Firmware:** FreeRTOS task affinity—Core 1 executes real-time sensor DSP; Core 0 handles Wi-Fi, encrypted local KLAP power queries, and Google Firebase RTDB cloud streaming.
- **Automated Force-ON Safety Guard:** Watchdog daemon detects accidental power switch-off within 1,000ms and immediately forces power back ON.
- **Centralized Township Interface:** Cloud-based facility management dashboard providing automated health triage maps and mobile repair alerts.

4. TARGET MARKET / BENEFICIARIES

Specific Industry or Sector: Commercial Facility Management (FM), Communal Places of Worship, Retail Real Estate, and Smart Sustainable Townships (e.g., Sime Darby Property's City of Elmina, Bandar Bukit Raja).

- **Communal Places of Worship:** Mosques, Suraus, and Kuils (temples) relying on central air-conditioning split units.
- **Commercial Centers & Clubhouses:** Community gyms, function halls, and retail malls (e.g., Elmina Lakeside Mall).
- **Local Maintenance Contractors (MSMEs):** Service technicians empowered with automated, predictive repair dispatching rather than reactive emergency calls.

Urban/Rural Focus: High-density master-planned urban townships and commercial property clusters.

5. UNIQUENESS AND INNOVATION

- **Ultra-Low Cost (< RM 50 Production BOM):** Industrial predictive systems cost RM 2,000–RM 5,000 per sensor. Apex brings industrial-grade prognostics to everyday township communal infrastructure at 1/40th the cost.
- **Solves the False Alarm Paradox:** By conditioning vibration and sound against ground-truth operational states, Apex prevents false emergency alarms during normal high-speed extractions.
- **Quantifies Parasitic Energy Drag in Ringgit:** Correlates vibration harmonics with active wattage to compute excess power loss (W_{excess}) and monthly electricity bill inflation.
- **Zero-Intrusive 5-Minute Retrofit:** Mounts externally with 3M VHB viscoelastic acrylic coupling and smart inline metering—zero rewiring or warranty-voiding alterations.

6. PROTOTYPE FEASIBILITY, EMPIRICAL EVIDENCE & THE EXPANSION PLAN

The Accessible Home Testbed (Proven Foundation): Because commercial 50 kW rooftop chillers are costly and inaccessible for early student bench testing, we built and validated our prototype on an accessible, highly dynamic residential front-load washing machine. Mechanically, a 1,400 RPM washing machine is one of the most violent rotating electromechanical environments in existence. By proving clean, synchronized dual-core telemetry here, we successfully de-risked the hardest technical hurdle.

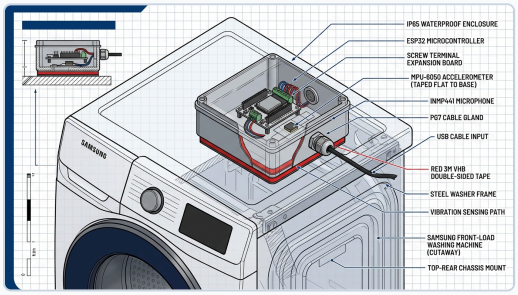


Figure 1: Field-Deployed Multimodal Edge Pod & Live Cloud Dashboard
• **Hardware Core:** ESP32 Dual-Core + MPU-6050 (±16g) + INMP441 I2S DMA + Tapo P110 (1 Hz KLAP).
• **Empirical Data in Hand:** Over **3,980 standardized active telemetry points** archived across 5 full wash cycles.
• **Live Global Dashboard:** Operational worldwide at **https://washer-pdm.pages.dev**.

The Bigger Expandable Plan: Because rotating electromechanical failure physics ($ ext{Vibration} ightarrow ext{Acoustic Friction} ightarrow ext{Electrical Drag} ightarrow ext{Seizure}$) is universal, our roadmap mass-produces these sub-RM 50 pods into clip-on hardware. We will deploy them across **20 to 50 commercial AC compressors in township mosques, kuils, community halls, and shopping malls**, training a specialized compressor ML model and providing facility managers with a centralized asset health dashboard.

7. EXPECTED IMPACT, SCALABILITY & COMMERCIAL ROI

Commercial Cluster Economics (20 Commercial AC Units):
• **Electricity Saved:** Eliminating 15% parasitic friction saves **~87,600 kWh annually (> RM 44,500/year** under TNB Tariff B).
• **Capital Payback Period: Under 1 month** against a sensor deployment cost of ~RM 3,400 (20 pods × RM 170 pilot kit).
• **Catastrophic Replacement Avoided:** Saves RM 15,000–RM 30,000 per prevented commercial compressor burnout.

Township Quality of Life & ESG Impact:

- **Zero Community Disruption:** Guarantees uninterrupted climate comfort during Friday prayers, temple festivals, and weekend retail crowds.
- **Circular Economy & ESG:** Extends equipment operating lifespan by **5 to 10 years**, preventing **25 to 40 metric tons of compressor scrap metal** and **66 metric tons of CO&sub2; emissions** per commercial cluster annually.